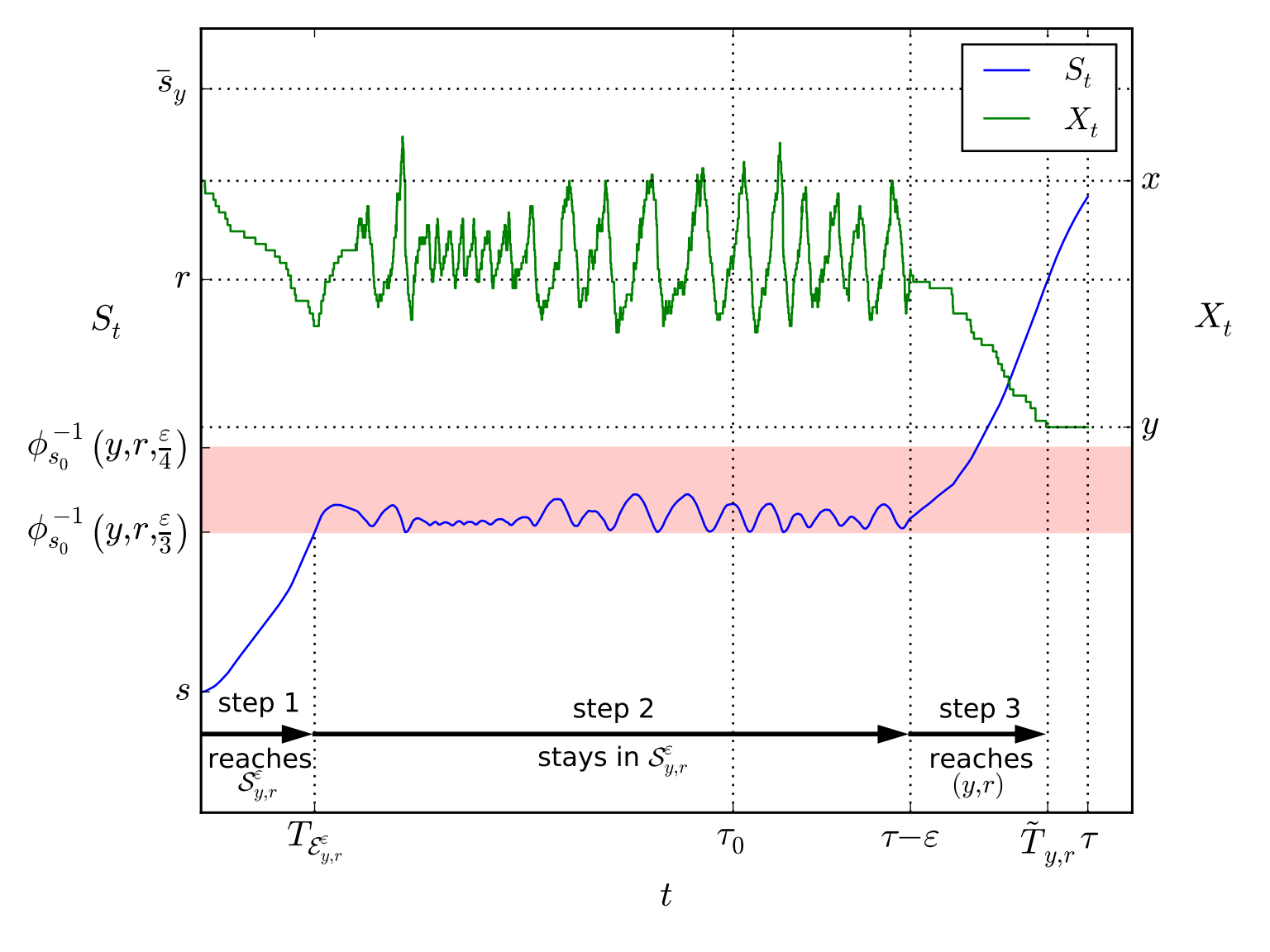
S_t
X_t
$\overline{s}_y$
r
x
y
$\phi_{s_0}^{-1}\left(y,r,\frac{\varepsilon}{4}\right)$
$\phi_{s_0}^{-1}\left(y,r,\frac{\varepsilon}{3}\right)$
s
step 1
reaches $\mathcal{S}_{y,r}^{\varepsilon}$
step 2
stays in $\mathcal{S}_{y,r}^{\varepsilon}$
step 3
reaches (y,r)
$T_{\mathcal{E}_{y,r}^{\varepsilon}}$
τ_0
$\tau-\varepsilon$
$\tilde{T}_{y,r}$
τ
t